



Maini Lotfi Abdelkader

Systems & Full Stack Developer

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Profile

A developer with experience in building web and mobile applications using React, Next.js, React Native, and Flutter. In addition to the modern web stack, I am a self taught systems programmer proficient in C and Rust. While I work across the full stack, my professional preference lies in Rust for its performance and memory safety. I am committed to writing clean, maintainable code and building modular tools from the user interface down to the system level.

Languages

Arabic – Native / Mother Tongue

Full command of reading, writing, and speaking; used in personal and professional communication.

French – Fluent, professional working proficiency

Able to communicate confidently in professional and technical contexts, including reading documentation and collaborating with French speaking colleagues.

English – Very good, strong written and spoken skills

Proficient in reading technical materials, writing code documentation, and communicating in professional or learning environments.

German – Currently learning, basic conversational level

Developing foundational reading, writing, and speaking skills to reach professional proficiency.

Education

MATHEMATICS BACCALAUREATE

Belkbire Abdelkader High School

- Completed secondary education with a focus on mathematics, developing strong analytical and problem-solving skills.

ATTENDED: TWO YEARS OF CIVIL ENGINEERING PROGRAM

Morsli Abdekah University

- Completed two years of civil engineering studies before transitioning to the tech field. Acquired foundational skills in technical reasoning, project planning, and logical problem solving, supporting a strong technical mindset for web development.

ADVANCED TECHNICIAN DIPLOMA IN WEB AND MOBILE DEVELOPMENT

National Institute of Specialized Vocational Training

- Specialized in full stack web and mobile development with hands on experience in HTML, CSS, JavaScript, React, Node.js, and PHP. Completed projects emphasizing responsive design, clean code, and user centered applications.

Skills

Systems & Languages:

Low-Level : Rust, C, Java

Web/Core : TypeScript, JavaScript (ES6+), SQL (CLI-Proficient) , PostgreSQL

Environments : Linux (Debian/Arch), Windows (WSL2) , Podman

Frontend & Mobile:

Frameworks : React, Next.js, React Native, Flutter

Visuals : Three.js, GSAP, Responsive Design

UI/UX : HTML5, CSS3, ShadCN UI, Tailwind CSS

Backend & Infrastructure:

Runtimes/Engines : Node.js, Express.js, PHP

Databases : MySQL, MongoDB, Supabase, Drizzle ORM

Services : REST APIs, Clerk (Auth), JWT, MinIO (S3), Zod

Development Workflow:

Editors : Neovim, Vim, VS Code

Version Control : Git, GitHub

Utilities : Bash/Zsh, npm, Postman, Chrome DevTools

Professional Experience

Freelance Web & Mobile Developer

Self employed / Algeria

- Developed responsive websites and mobile applications for clients, using HTML, CSS, JavaScript, React, Node.js, and PHP.
- Created and maintained online stores and business websites, ensuring user-friendly design, functionality, and performance.
- Collaborated with clients to understand requirements, deliver projects on time, and implement iterative improvements.

Freelance Graphic Designer & 3D Modeling Artist

Self employed / Algeria

- Designed logos, banners, and visual content for web and print, meeting client branding requirements.
- Created 3D models and renders using Blender for various projects, including product visualization and creative presentations.

Systems Projects

Linux System Monitor (C Systems)

Low Level Kernel Telemetry Tool

Github 

(C, Linux Kernel, Procfs, ANSI/VT100)

- Engineered a high efficiency system monitor in C that interfaces directly with the Linux /proc virtual filesystem to extract real time hardware telemetry without external library overhead.
- Implemented a differential CPU analysis engine to calculate utilization metrics by processing kernel clock ticks and managing state deltas across high precision timed intervals.
- Optimized memory footprint and I/O performance by utilizing low level stream parsing and ANSI escape sequences to deliver a flicker free dashboard with near zero RAM allocation.

Msh

UNIX Compatible System Shell

Github 

(C, POSIX API, GNU Libc)

- Engineered a robust command-line interpreter featuring a custom recursive-descent parser and a multi-stage execution engine to facilitate seamless interaction between the user and the UNIX kernel.
- Architected a parallel execution pipeline utilizing pipe(), fork(), and file descriptor redirection (dup2) to support multi-stage command streaming with proper process synchronization and zombie process reaping.
- Developed a dynamic memory management system for command tokenization and environment variable handling, ensuring stable buffer resizing and leak-free operation during high-frequency I/O operations.
- Implemented a modular builtin command suite and a persistent execution loop with terminal state awareness, providing a foundation for advanced shell features like command history and tab completion.

Rust TUI System Monitor

High Performance Terminal Telemetry Dashboard

Github 

(Rust, Ratatui, Sysinfo, Crossterm)

- Engineered a stateful Terminal User Interface (TUI) using an immediate mode rendering engine to provide real time visibility into CPU, RAM, and network throughput with sub millisecond latency.
- Implemented a non blocking event polling system and a persistent tracker struct to calculate network deltas and dynamic process prioritization without interrupting the 500ms refresh cycle.
- Developed a memory safe hardware abstraction layer using Rust's ownership model to manage terminal raw mode, alternate screen buffers, and graceful system restoration upon application exit.

Echo

Multi-tenant SaaS Storage Engine & IAM Framework

Github 

(Node.js, TypeScript, Drizzle ORM, MinIO/S3, Podman)

- Architected a modular monolith using a Service Repository pattern to provide strict data isolation across a multi tenant hierarchy, managing complex relational memberships between Users and Organizations.
- Engineered an S3 compatible object storage layer utilizing Multer memory-storage to stream binary data directly to a containerized MinIO instance, bypassing local filesystem persistence to ensure horizontal scalability.
- Implemented a secure asset retrieval system leveraging time limited cryptographic presigned URLs and Version 4 UUID key obfuscation to prevent unauthorized access and physical filename exposure.
- Developed a type safe metadata tracking engine with Drizzle ORM and PostgreSQL to manage MIME type validation, file ownership, and organizational partitioning via foreign key constraints.
- Optimized local first development infrastructure on Linux using Podman native containerization and SELinux labeled volumes to maintain environmental parity with production cloud storage environments.

Full Stack Projects

Valkyrie

Distributed Load-Testing Framework

GitHUB 

(Node.js, Redis Streams, TypeScript, Blessed-Contrib)

- Engineered a distributed Controller-Worker architecture utilizing a Redis backbone (Pub/Sub and Streams) to orchestrate high-concurrency traffic clusters with sub-millisecond command synchronization.
- Developed a real-time Telemetry Ingestion Engine that processes high-throughput metric streams and calculates P99 latency distributions and RPS (Requests Per Second) via a persistent Redis Consumer Group.
- Architected a low-latency TUI Dashboard featuring dynamic line graphs and node topology tables, optimized with Smart Control Sequence Rom (CSR) to minimize CPU overhead during millisecond UI refreshes.
- Implemented a stateless horizontal scaling protocol allowing for live injection of target URLs and aggressivity multipliers across the mesh without interrupting active worker processes.

Kaen

Real time Community Platform for Developers

Github 

(React, TypeScript, Supabase, TanStack Query)

- Built a high-performance discussion engine featuring real-time post synchronization via PostgreSQL replication, an instant global search with optimized filtering, and a recursive nested comment architecture for complex threading.
- Implemented a robust engagement system including a debounced upvoting/liking mechanism to minimize API overhead, dynamic comment counts, and granular data security through Supabase Row Level Security (RLS).
- Designed a responsive "HUD-style" interface utilizing Framer Motion's hardware-accelerated transitions, featuring adaptive Grid/List view variants and mobile-first metadata optimization.

Archfolio

Operating System Interactive Portfolio

Github 

(React, GSAP, Tailwind CSS, JavaScript)

- Built a full-scale virtual desktop environment featuring a window management system, draggable GUI components, and a custom-built file system architecture.
- Implemented a dual-mode navigation engine including a functional terminal emulator with command-line parsing and a graphical file manager for directory traversal.
- Designed a high-performance animation system using GSAP for window logic and state-driven UI transitions, ensuring a seamless user experience across modular components.

Vex

AI Tutor SaaS Platform

Github 

(Next.js, TypeScript, Supabase, Stripe)

- Built a full-stack SaaS platform that allows users to create AI tutors and chat with them in real time.
- Implemented secure authentication, protected routes, and subscription-based payments using Stripe.
- Designed a scalable architecture with modular components, server/client separation, and responsive UI using Tailwind CS

Estus Engine

Distributed Task Orchestration & Background Worker System

Github [🔗](#)

(Node.js, TypeScript, BullMQ, Redis, Podman)

- Architected a distributed producer consumer engine to offload heavy workloads from the main execution thread, enabling a non-blocking request-response lifecycle and increased system throughput.
- Implemented a persistent message broker using containerized Redis to manage job states and high-concurrency queuing with zero data loss across system restarts.
- Engineered a resilient task lifecycle manager with BullMQ, incorporating exponential backoff, automatic retries, and stalled-job detection to ensure integrity under peak loads.
- Developed a type safe telemetry dashboard using React and Tailwind CSS v4, providing real-time visibility into job dispatching via a unified TypeScript interface layer.
- Optimized environment orchestration via Podman containerization, ensuring architectural parity between Fedora development environments and production Linux clusters.

Sven

Real time Language Exchange Platform

Github [🔗](#)

(React, Node.js, Express.js, React Query, Axios)

- Engineered a real-time communication platform facilitating interactive language exchange through live chat and video call integration, focusing on low-latency data synchronization.
- Developed a robust state management layer using React Query and Axios to handle asynchronous data fetching and real-time UI updates across complex learning workflows.
- Architected a scalable RESTful backend to manage user authentication, immersive conversation sessions, and structured practice modules for a seamless peer-to-peer experience.